

B Rakeshkumar

Embedded Systems Engineer · ECE Undergraduate (Graduating 2027)

Mangaluru, Karnataka, India · rakeshbhandari041@gmail.com · +91 9481288157

linkedin.com/in/b-rakesh-kumar · github.com/Rakesh-Bhandari

SUMMARY

Embedded Systems Engineer (fresher) with hands-on experience in real-time sensor systems (I2C/SPI/UART), RF communication protocol design (NRF24L01), and 4-layer custom PCB design in KiCad. Comfortable across the full embedded stack: schematic capture, hardware bring-up, and firmware in Embedded C and PlatformIO.

EDUCATION

B.E. Electronics and Communication, Shree Devi Institute of Technology, Mangaluru (VTU) · 2023-2027 · CGPA 8.45/10

Pre-University Course (PCMCs), Government PU College · 2021-2023 · 82.22%

SSLC, Pragathi Vidyalaya English Medium High School, Muroor · 2021 · 86.77%

PROJECTS

- Custom ESP32-S3 Development Board: 4-layer KiCad design, 3 hierarchical sheets, BOM, power path, DRC-verified.
- Emergency Vehicle Priority System (V2X): 3-node ESP32 + NRF24L01 + IR system; RF packet protocol and state machine.
- IoT Smart Bridge Health Monitoring: ESP32, MPU6050 (I2C), flex and water sensors; React/Flask/MySQL dashboard; cron jobs.

EXPERIENCE & ACHIEVEMENTS

- 1st Place, Blind Coding, Envision 2026 (national techno-cultural fest) · April 2026
- Sankalp'25 24-hour National Hackathon (IEEE) · Team shortlisted · Smart Parking System hardware/firmware
- Code Meet 2025 Hackathon · Frontend and AI chatbot module for Learnly (team of 4)

SKILLS

Languages: C/C++, Embedded C, Python (basics), HTML & CSS

Protocols: I2C, SPI, UART, RF (NRF24L01), IR

Tools: Arduino IDE, PlatformIO, STM32CubeIDE, KiCad, VS Code, Git & GitHub, Linux (basics)

Hardware: Digital Electronics, Circuit Design & Debugging, Sensor Interfacing, Multi-layer PCB Design, Soldering

CERTIFICATIONS

Embedded System Design with ARM (NPTEL) · Introduction to RISC-V (Linux Foundation) · Git & GitHub Bootcamp (LetsU)